

Chapter 1

Web Technology

INT150 Web Technologies (2/2025)

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Topics

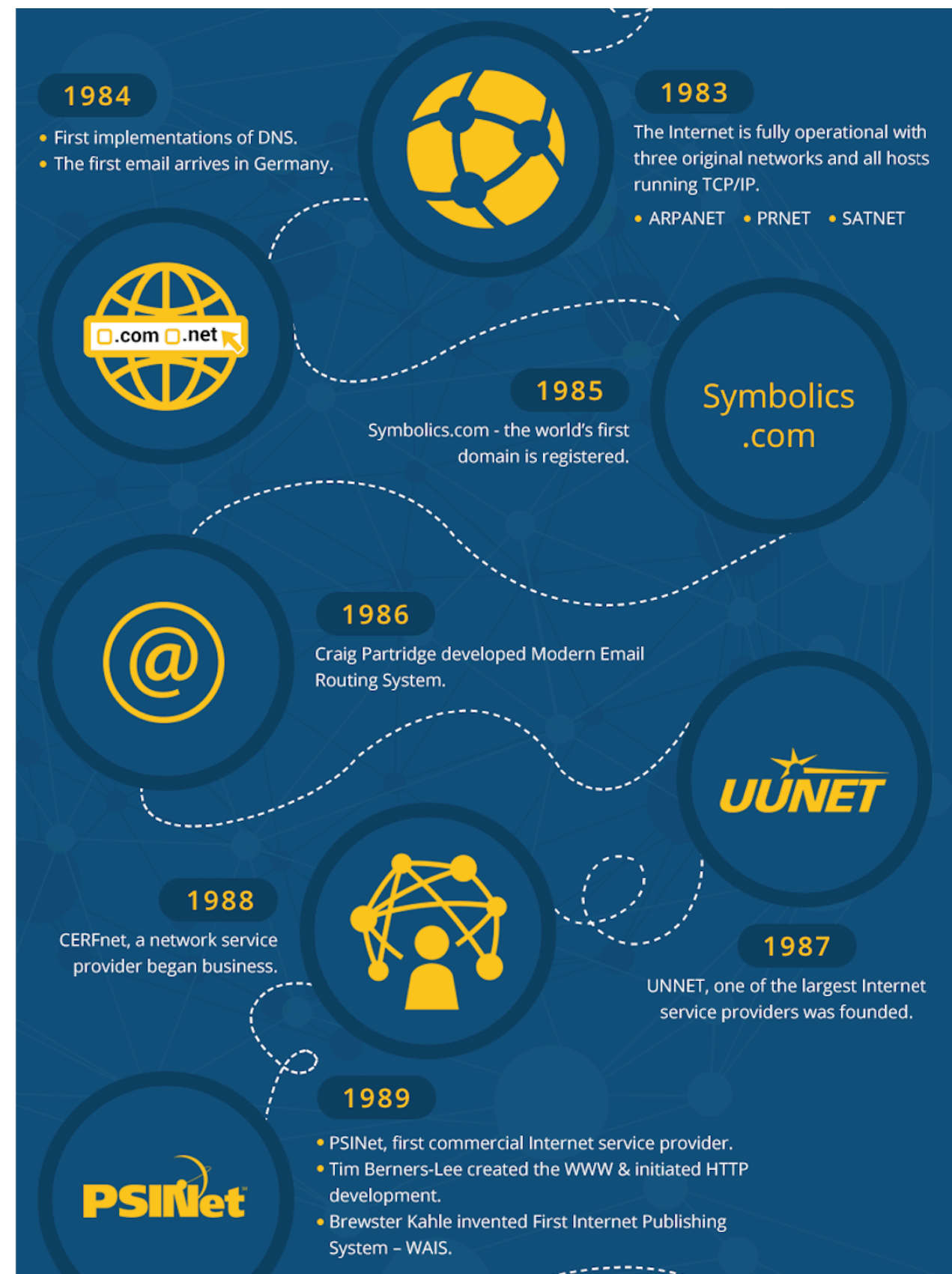
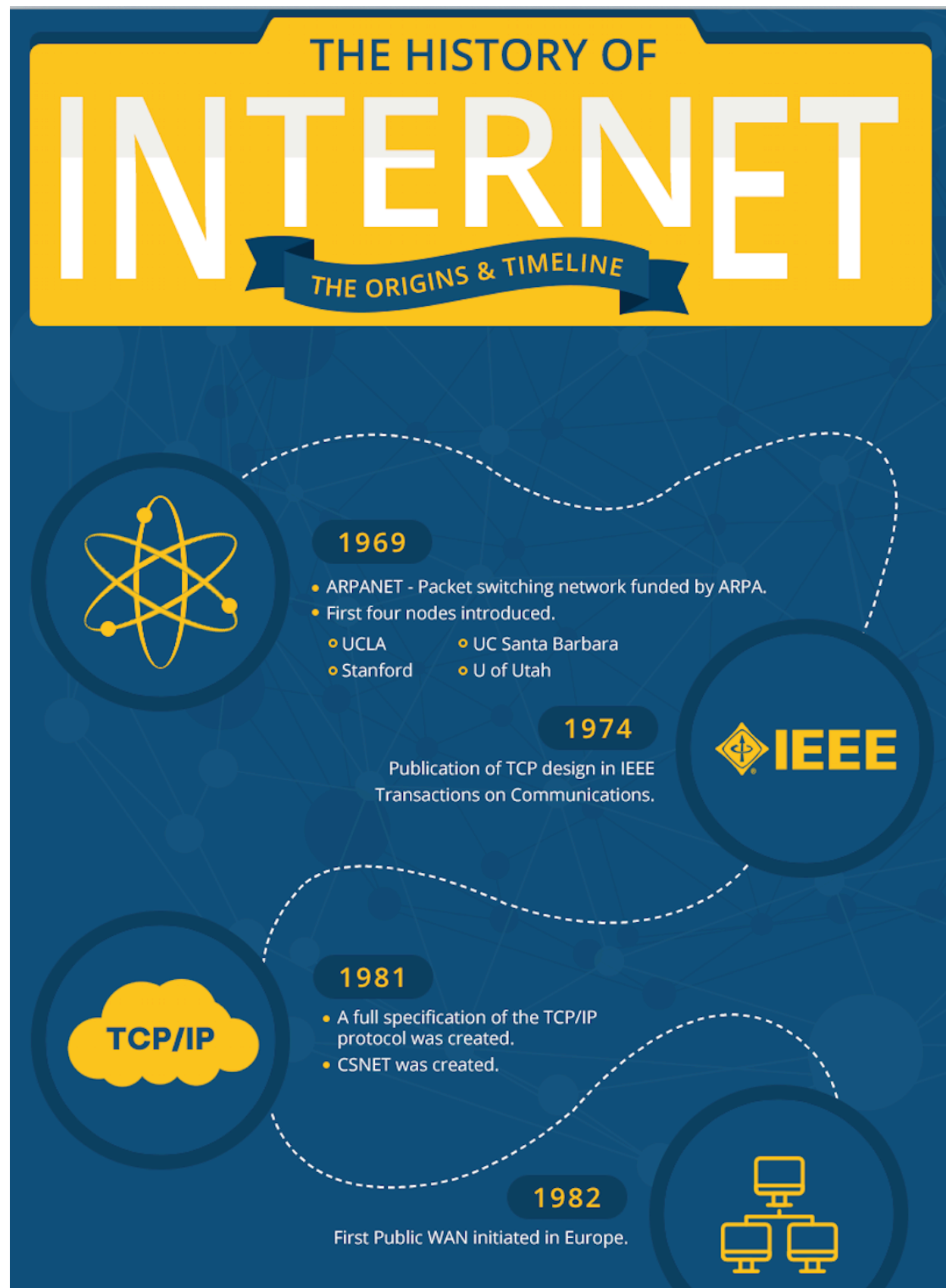
- What will you learn after finishing this class:
 - The Internet
 - Evolution of the Internet
 - World Wide Web (www)
 - Web Browser and Web Page Address
 - Web Sites and Web Pages

The Internet

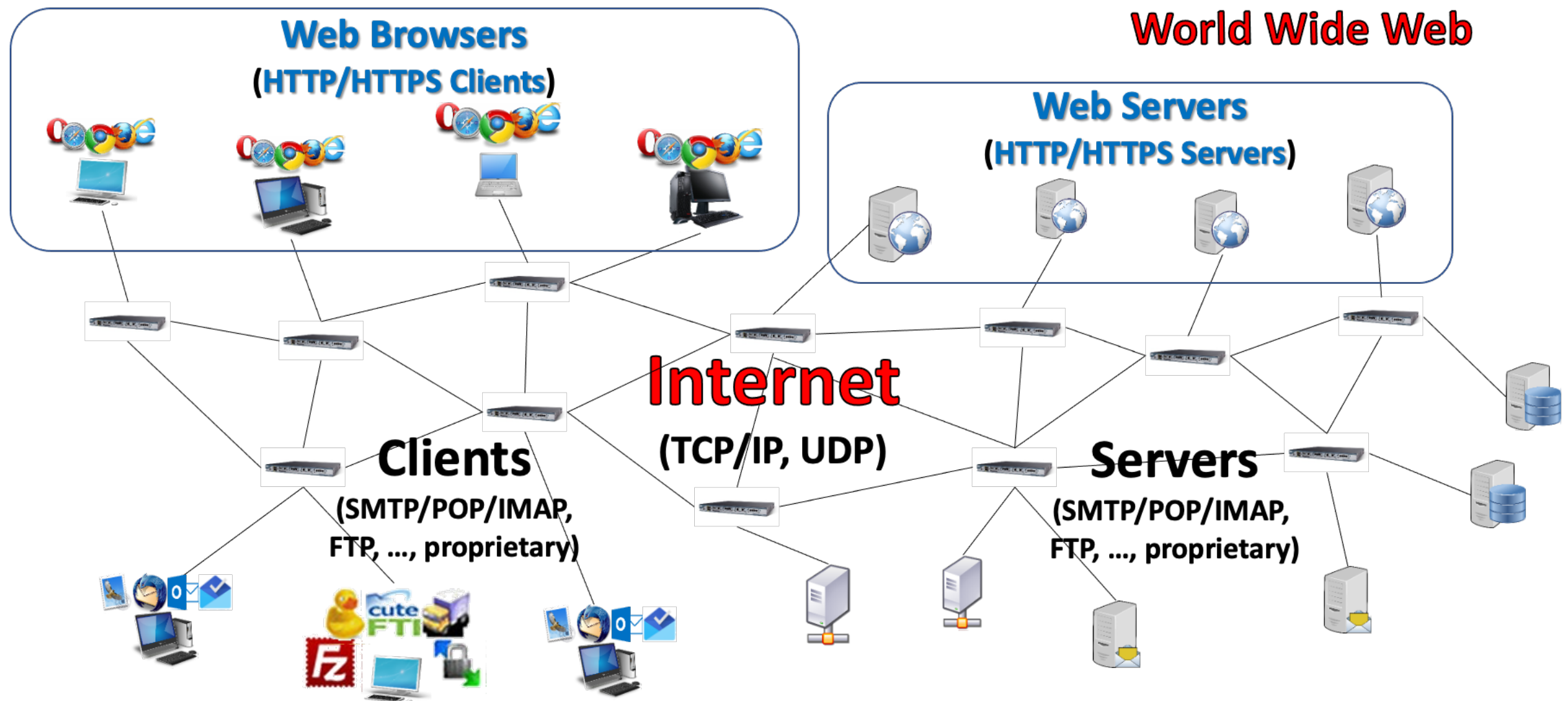
- Is a world-wide collection of **networks** that links millions of business, government agencies, educational institutions and individuals.
- The **purpose** of connecting computers together is to **share** information.
- Many ways to share information through:
 - World Wide Web (Web)
 - Email
 - File Transfer
 - Chat (Messenger, Line, Discord)
 - Social Media (Facebook, Twitter, Instagram)
- These standardized methods for transferring data or documents over a network are known as **protocols**

Evolution of the Internet

- A project of the U.S. Department of Defense's [ARPA](#) ([ARPANET](#)) in the late 1969s.
- [ARPA](#)'s goal was to build a network that
 - allow scientists at different locations to [share](#) information and work together
 - could [function](#) even if part of the network were disabled or destroyed by a disaster such as nuclear attack.
- [ARPANET](#) originally consisted of [four](#) main computers .
- By 1984, [ARPANET](#) has more than 1,000 individual computers linked as hosts.
- Today, more than 550 million hosts connect to the Internet.



Internet and World Wide Web



Web

- [Web](#) or [World Wide Web](#) ([www](#))
 - is one of the ways information can be shared over the internet.
 - consists of a worldwide collection of electronic documents ([Web pages](#))
- It allows documents ([Web pages](#)) to be linked to one another using [hypertext](#) links.
- The Web uses a protocol called HyperText Transfer Protocol ([HTTP](#)).

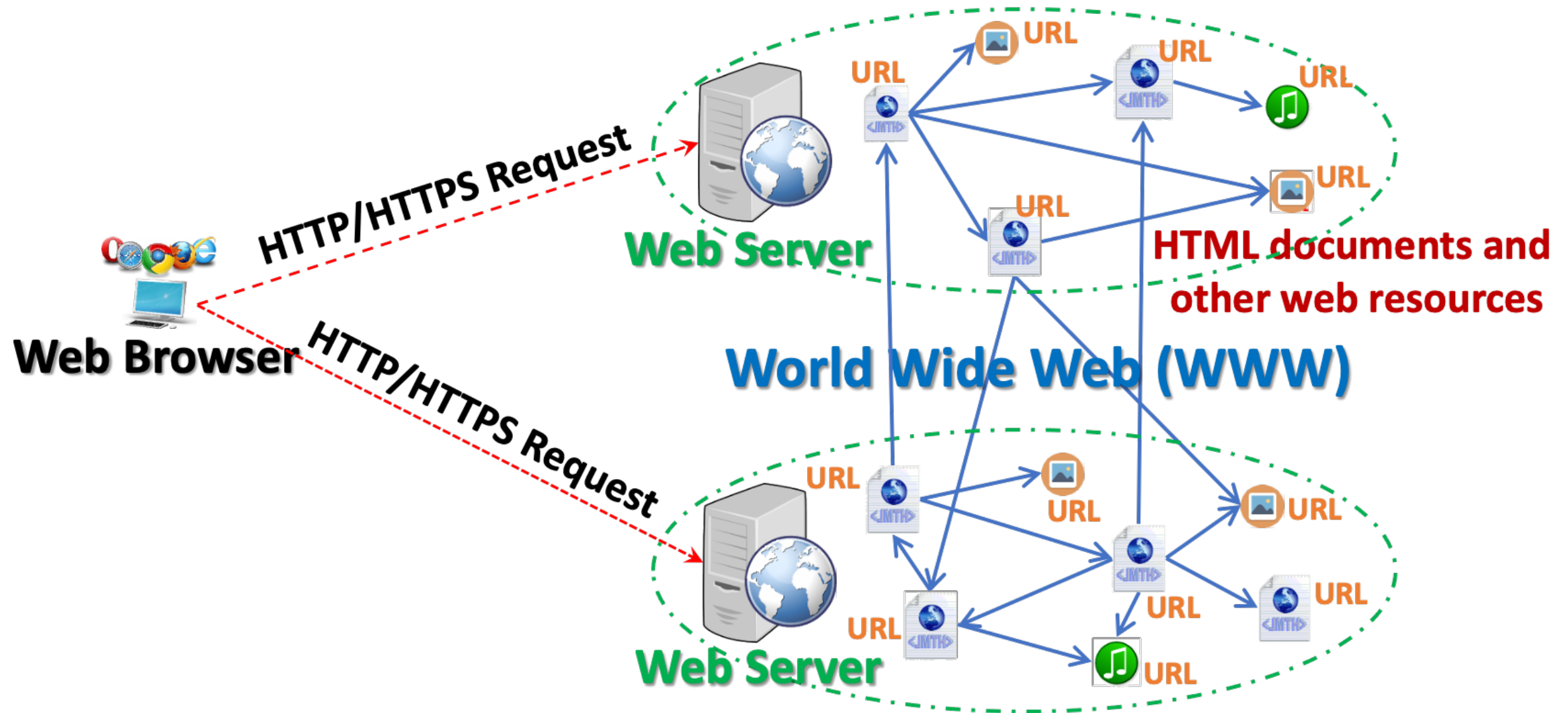
Server

- The computers make up the Internet
 - They "serve up" documents upon request
 - These computers are known as [Servers](#).
- The [server](#)
 - is the [software](#) that allows the computer to communicate with other computers.
 - is to wait for a request for information, then retrieve and send that information back to requesters ([Clients](#)).
 - is known as [Web Server](#) or [HTTP Server](#).
 - Two most popular web servers : [Apache](#) and Microsoft Internet Information Services ([IIS](#))
 - Today, most popular web servers : [Nginx](#), Apache, [LiteSpeed](#), Microsoft IIS and [Node.js](#).

(Web) Browser

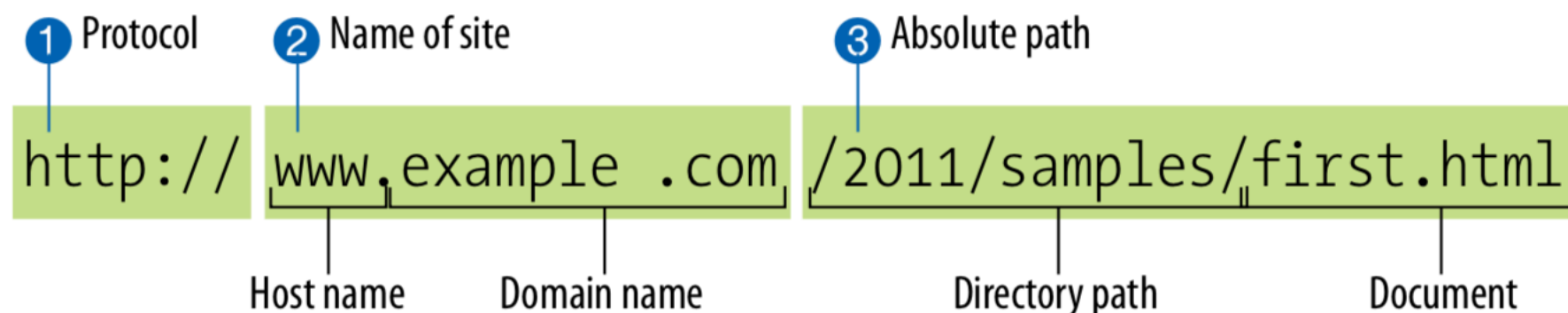
- Is [application software](#) that allows users to access and view Web pages.
- People use desktop browsers, mobile browsers and other assistive technologies (such as screen readers) as clients to access documents on the Web.
- The [requests](#) and [responses](#) are handled via [HTTP](#) protocol.
- [HTTP](#) can use to transfer images, movies, audio files, data, scripts and all the other web resources.
- The browser as a window on a computer monitor with a web page displayed in it
- The most popular desktop browsers:
 - [Internet Explorer](#) (Windows), [Chrome](#), [Firefox](#), [Safari](#) (Mac)

HTML & HTTP/HTTPS



Web Page Address (URLs)

- Every page and resource on the web has its own special address called a Uniform Resource Locator (URL).
- The parts of URL:



- When you enter the URL in the address bar in Browsers, you can omit the <http://> part. Browsers will add <http://> automatically as a convenience to save you some keystrokes.

IP Address

- Every computer and device (modem, router, smartphone, car, etc) connected to the Internet is assigned a unique numeric [IP address \(Internet Protocol Address\)](#).
- The server that hosts www.sit.kmutt.ac.th has IP address 202.44.9.116
- The IP address usually consists of four groups of numbers, each separated by a period. The number in each group is between 0 and 255.
- This format of IP address is called [IPv4](#). It produces 4.3 billion unique addresses.

IPv6

- The IANA, the organization that assigns IP numbers, assigned the last bundle of IP addresses on February 3, 2011.
- There is no more ###.###.###.###-style IPs.
- The solution is a new IP format (IPv6) that allows for trillions and trillions of unique IP numbers.
- IPv6 will operate parallel to IPv4 today.
- IPv4 will be phased out in decades.

Domain Name System (DNS)

- The **IP addresses** are difficult to remember and use (by human).
- The Internet supports to the use of text name that represents one or more IP addresses.
- A **domain name** is the text version of an **IP address**.
- The components of a **domain name** are separated by periods.
- The **domain name system (DNS)** is the method that the Internet uses to store domain names and their corresponding IP addresses.
- When you specify a **domain name**, a **DNS** server translates the domain name to its associated ID address.

Domain Name => www.sit.kmutt.ac.th

IP address => 202.44.9.116

Top-level Domain (TLD)

- Is the last section of the domain name
- A generic TLD (gTLD) such as com identifies the type of organization associated with the domain
- The organization that assigns and controls top-level domains is the [Internet Corporation for Assigned Names and Numbers \(ICANN](#) pronounced EYE-can).
- **Ex.** biz, com, info, name, net, and org
- For [international Web sites](#) outside the US, the domain name also includes a country code TLD (ccTLD)
- **Ex** com.au, ac.th, co.th

Generic Top-Level Domains

gTLD	Intended Purpose
biz	Business of all sizes
com	Commercial organizations, businesses, and companies
edu	Educational institutions
gov	Government agencies
mil	Military organizations
name	Individuals or families
net	Network providers or commercial companies
org	Nonprofit organizations

Web Site

- Is a collection of related [Web pages](#) and associated items, such as documents and pictures, stored on a [Web server](#).
- The same Web server can store multiple Web sites.
- [Home page](#) refers to the first page that a Web site displays.
- The [home page](#) provides information about the Web site's purpose and content.

HOW TO

CREATE A WEBSITE



HOW WEBSITES WORK:

A HOUSING ANALOGY

WEB HOSTING IS LIKE A HOUSE.

Most people rent house(s) or room(s) (subscribe to hosting packages) to stay (host their website).

WEBSITE IS EVERYTHING IN YOUR HOUSE:

furniture, interior design, etc. Visitors might leave your house if it's stinky and ugly! (Badly designed website)

If you have a lot of **VISITORS**, you need rent more space to fit in more people.

DOMAIN NAME IS LIKE A STREET ADDRESS.

You need to register one so that people can find you.

INTERNET IS LIKE HIGHWAYS AND ROADS. WEB BROWSER IS LIKE A CAR.

You enter the domain name and it will bring you to the destination. Just like cars, some browsers are fast and some are really slow.

CHROME

INTERNET EXPLORER

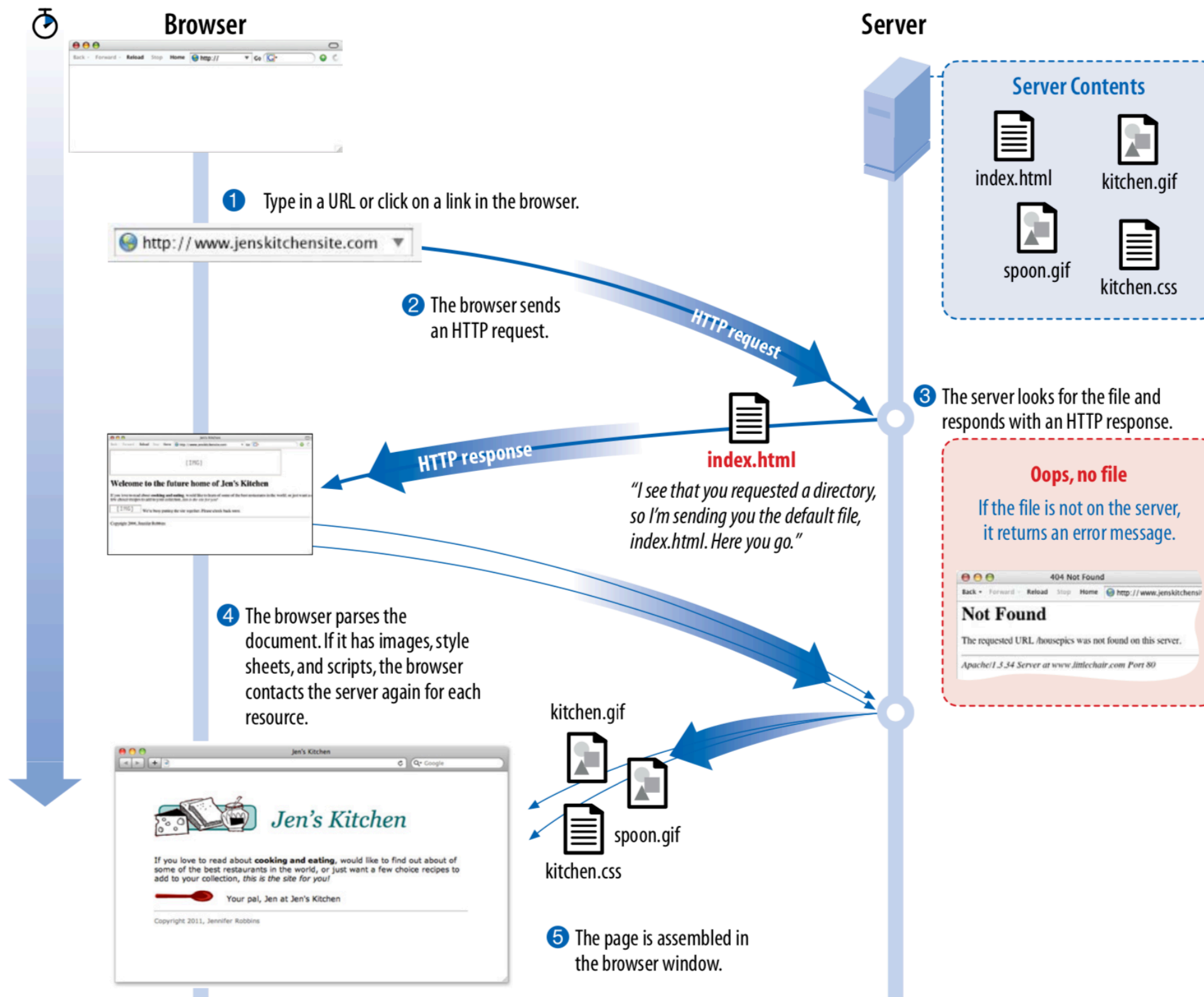


www.domain.com

Web Page

- is a document commonly written in [Hypertext Markup Language \(HTML\)](#) or HTML document.
- is accessible through the Internet or other networks using Web browser.
- It may contain text, graphics and hyperlinks to other web pages and files.

How browsers display web pages



Client-side

Client-side: Web Browser

- **Hypertext Markup Language (HTML)**

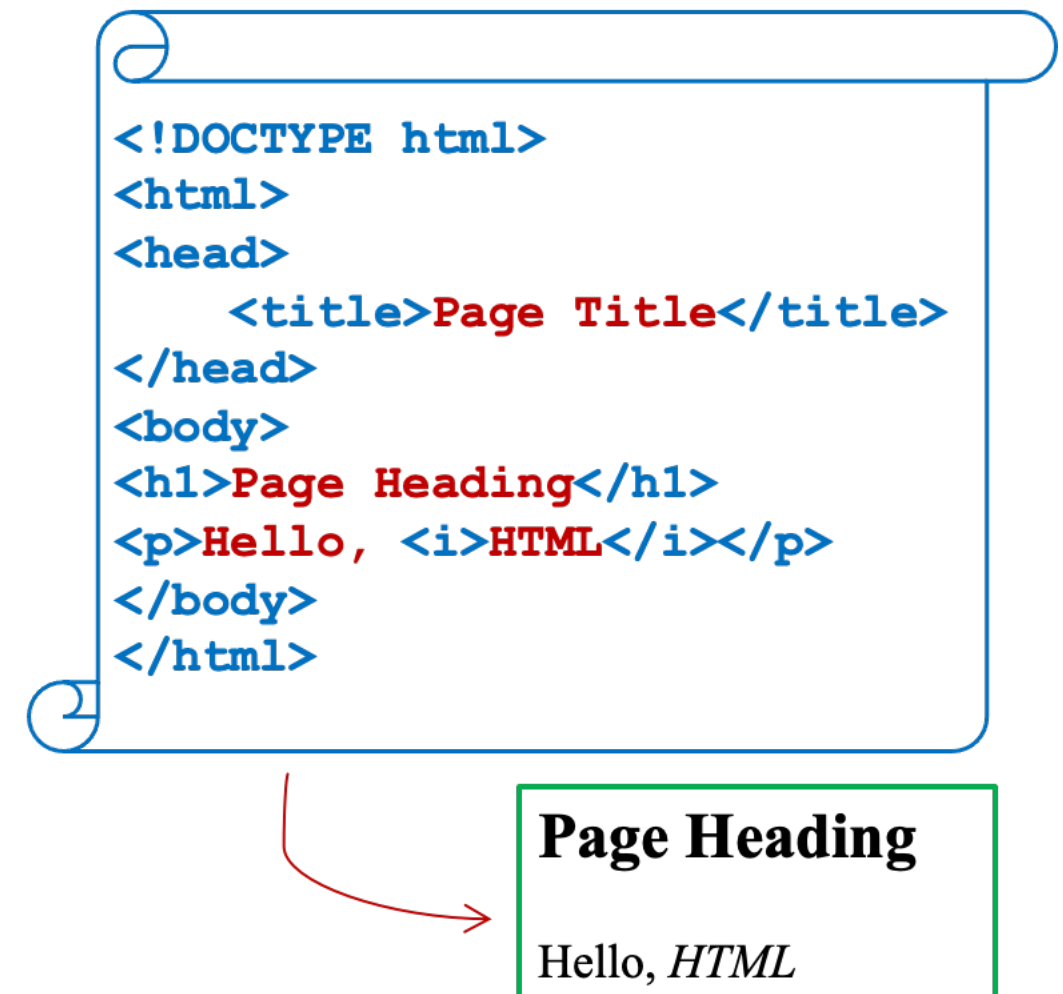
- the standard markup language for documents designed to be displayed in a web browser.
- interlinks web documents/resources to enable users to move from one document to another.

- **Cascading Stylesheet (CSS)**

- for describing the presentation of a document written in a markup language like HTML.
- enables **the separation of presentation and content**, which improves content accessibility and provides more flexibility and control in the specification of presentation characteristics.

- **Client-side JavaScript (JS, ECMAScript)**

- enables interactive web pages.



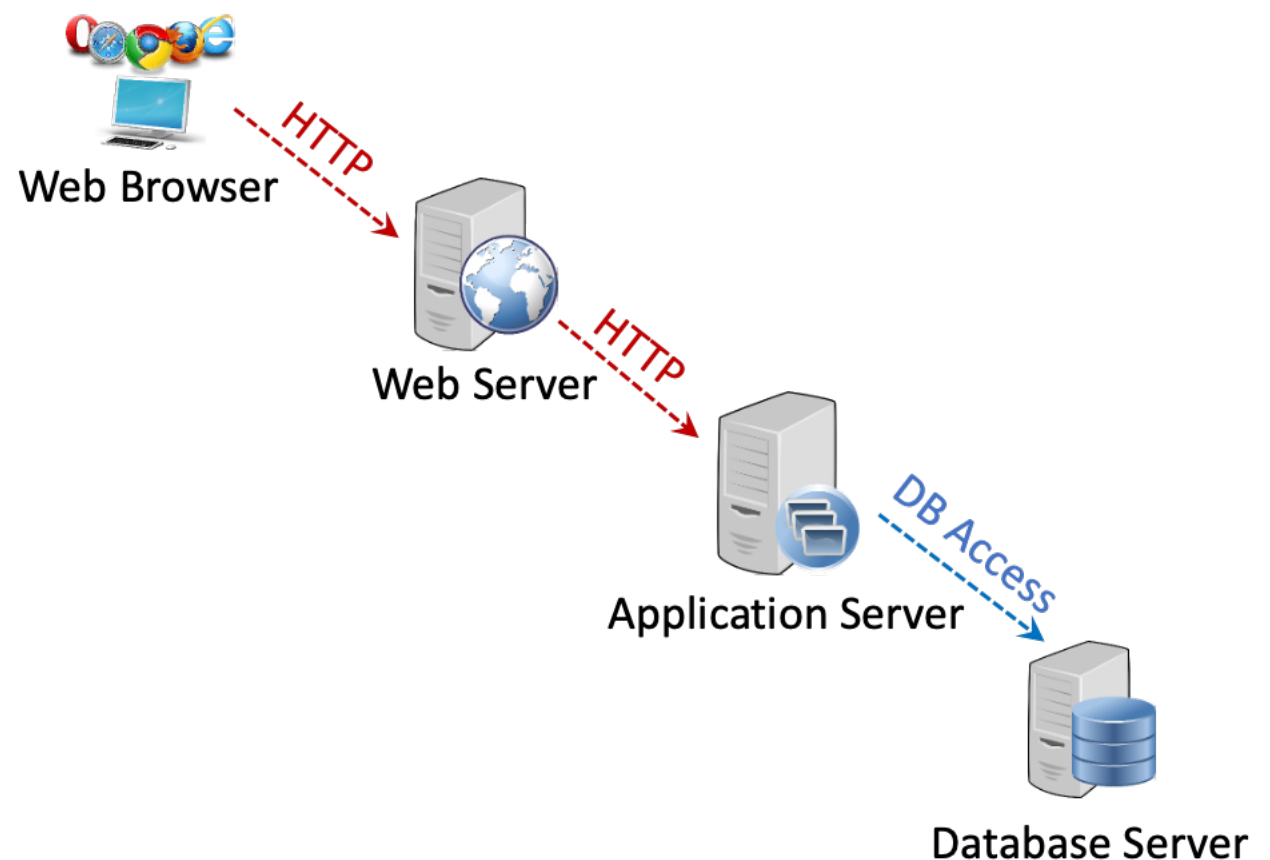
Server-side: Web Server and Web Application Server

- **Web Server**

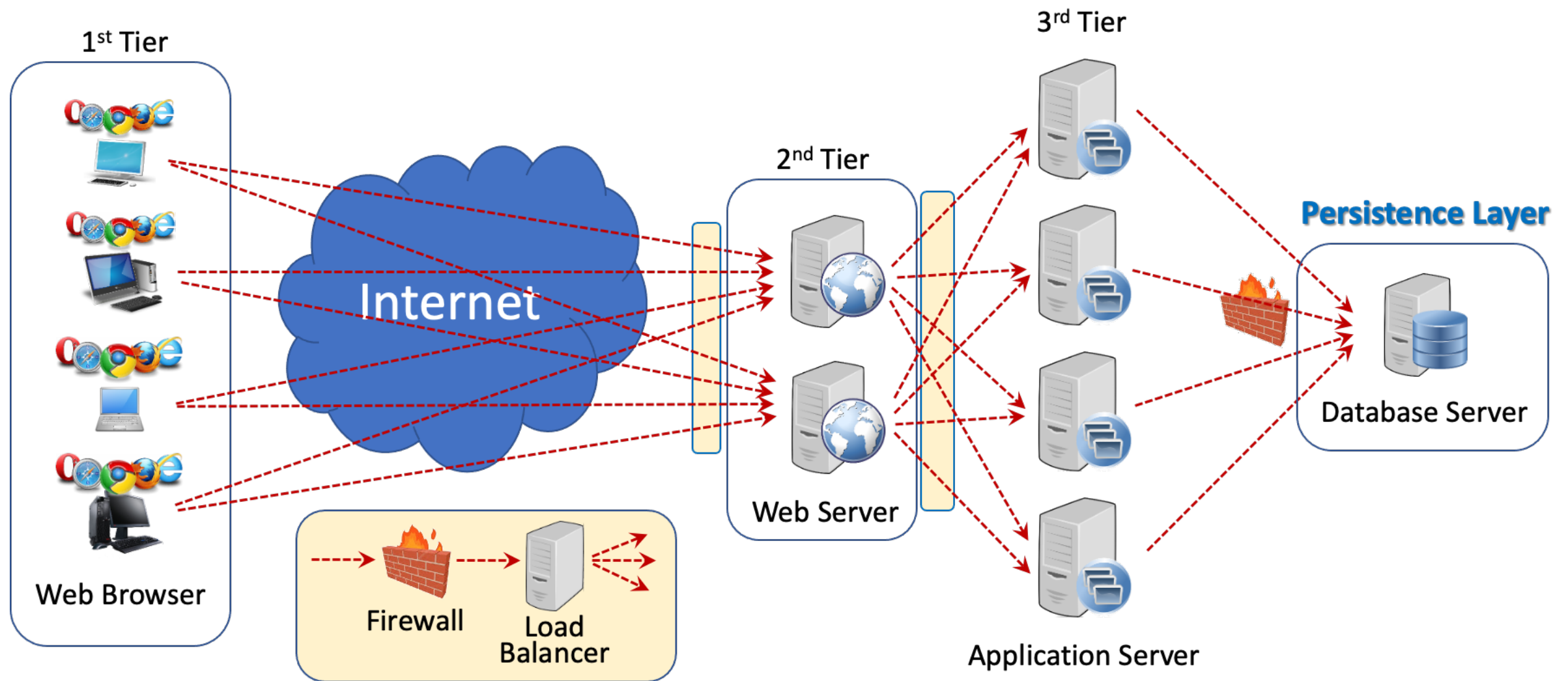
- Serve HTTP/HTTPS Requests.
- Mostly serve static documents; or forward the requests to Application Servers.

- **(Web) Application Server**

- Process user requests forwarded from web servers.
- Usually have a DBMS behind it for storing/retrieving data.



Three-Tier / N-Tier Architecture



Web Application Layer Architecture

Separation of Concerns

- Layer Architecture
 - Presentation Layer
 - Responsible for **User Interface**, Input Validation, UX/UI (User Experience/ User Interface)
 - Business Logic Layer
 - Responsible for **business logic of the application**
 - Persistence Layer
 - Responsible for **persistence of data**

Framework and Environment for Web Application

- Java
 - Servlet/JSP, Spring, EJB
- .Net Framework
 - ASP
- PHP
 - Laravel, CodeIgniter, CakePHP
- JavaScript
 - Node.js, Express.js, AdonisJS
- Python
 - Django



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